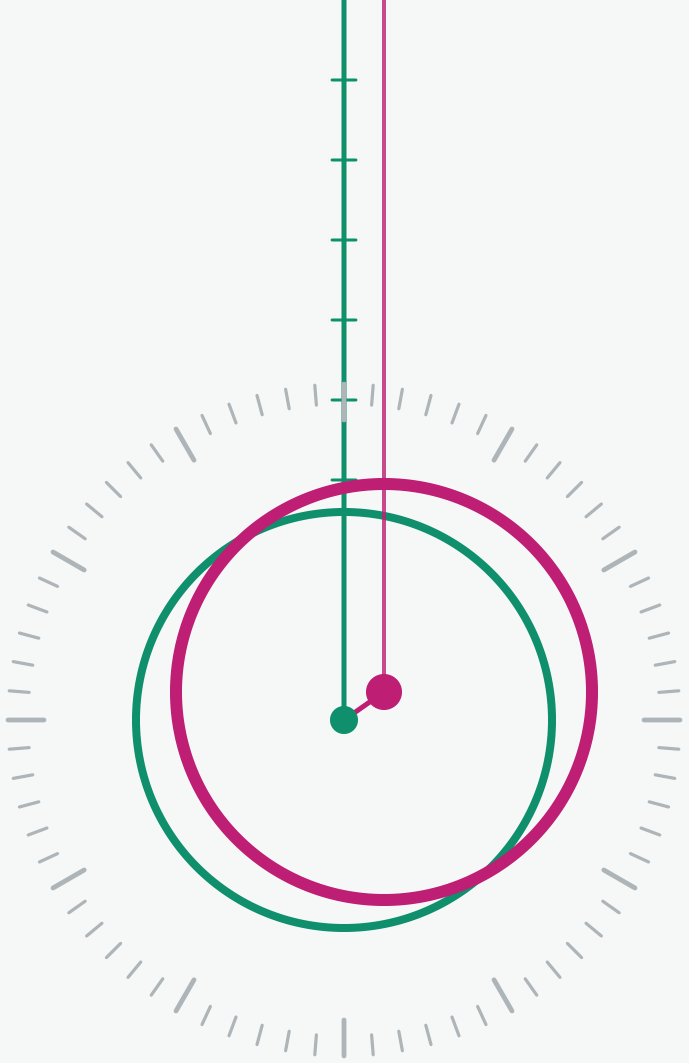
DATUM LINE

AI × BIO

Confidence with controls.

A shared map for every lab’s cells.



Kat Titterton · San Francisco



macrofluidic.me/resume

TEAM

one founder · looking for a technical co-founder

collaborators in industry

Katherine Titterton

five labs · one **unanswered question**

2023–2026 — Axiom SAN FRANCISCO
founding MTS · hepatocytes at scale

2022–2023 — insitro SOUTH SAN FRANCISCO
neuroscience assays · productionization

2021–2022 — MIT SYNTHETIC NEUROBIOLOGY
expansion microscopy

2018–2021 — abbvie NEUROSCIENCE
tau · neurons

2017–2018 — MGH HARVARD MEDICAL SCHOOL
Xenopus oocytes · $GABA_A$ receptors

2017 — COLGATE BA
neuroscience + math biology

MY WORK

1. Cell Painting in primary human hepatocytes
   Cell Systems 2026 · Axiom Bio
2. Counting cells predicts bioactivity benchmarks
   Nature Communications 2026 · Axiom Bio
3. A neuronal tau aggregation assay
   J Biol Chem 2025 · AbbVie
4. Multiplexed expansion revealing
   Nature Communications 2024 · MIT
5. Neurosteroid photolabeling of $GABA_A$ receptors
   Eur J Med Chem 2019 · MGH · Harvard Medical School

PUBLISHED WITH

Broad Institute · MIT · AbbVie · EMBL-EBI · HESI · Bayer · University of Cambridge · Uppsala University

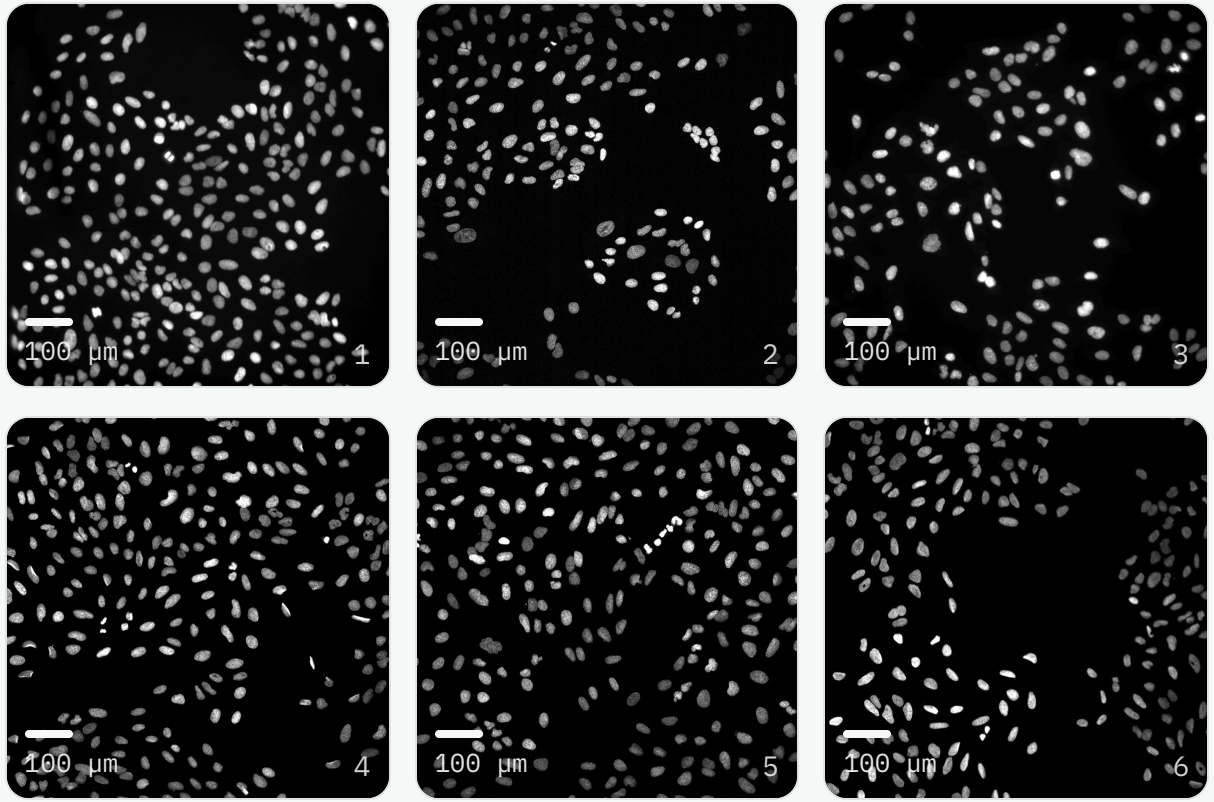
THE QUESTION

“Do my cells look normal?”

asked at five benches · answered at none

One lab, or six?

Cell Painting six dyes, 3,773 numbers per well



same line ✓

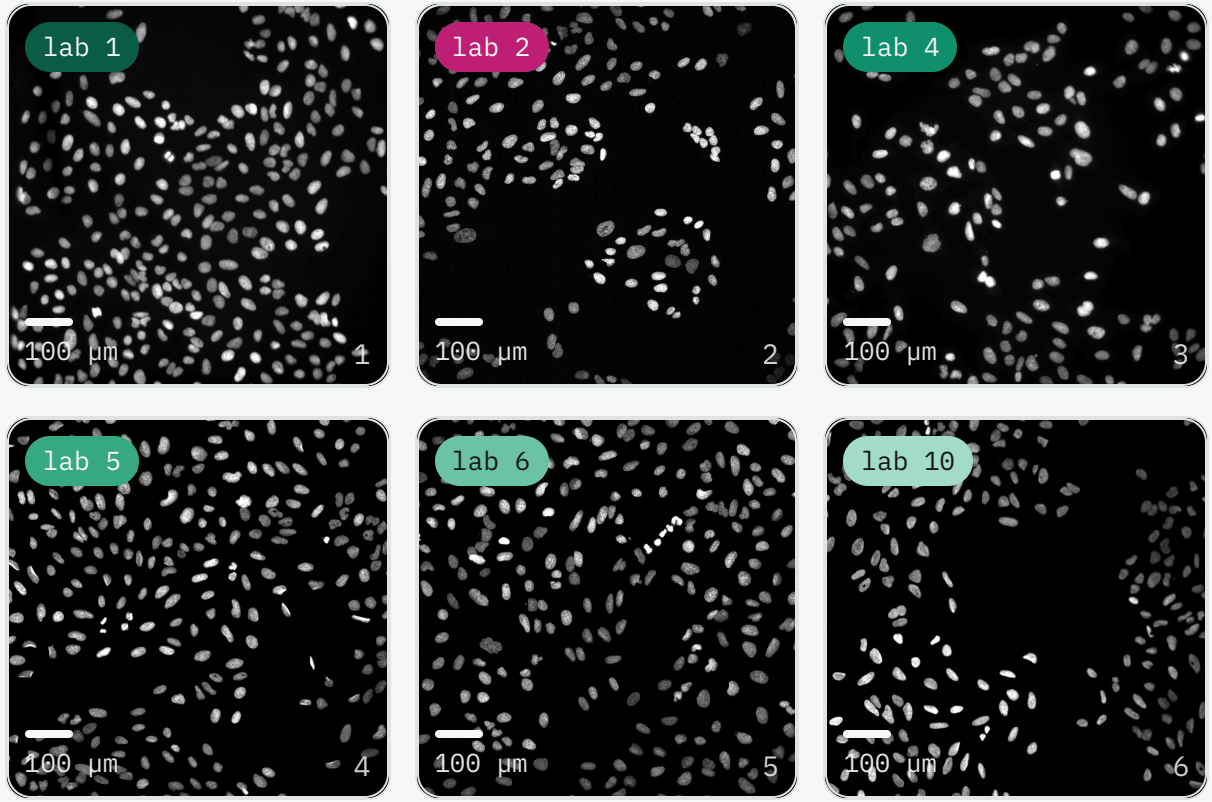
same protocol ✓

no drug ✓

fig 1 93,228 wells · 11 labs · plates held out · JUMP [7]

click → the answer

Could you tell? A classifier can.



normalised to itself → 24% · A3

fig 1 93,228 wells · 11 labs · plates held out · JUMP [7]

Cell Painting six dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE, 1 IN 11 LABS

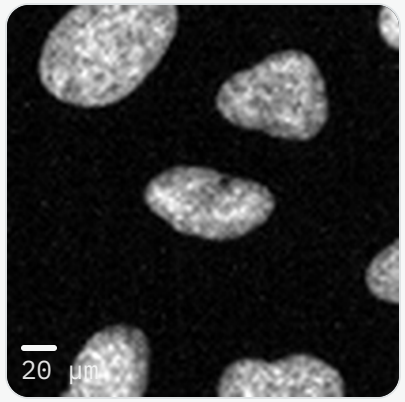
78 wells wrong, of 93,228

click → follow one well

One well, all the way out.

Cell Painting six dyes, 3,773 numbers per well

ONE NUCLEUS



192 µm² · 0.59

ITS WELL

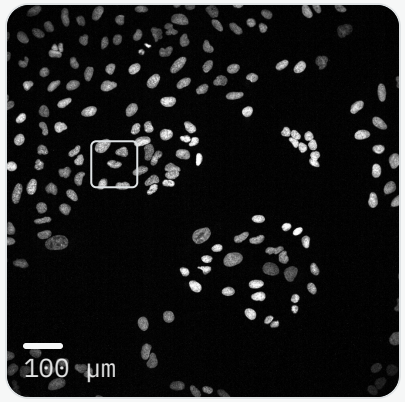


plate 1053600681

3,773 numbers → one point

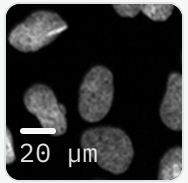
WHERE IT LANDS



93,228 wells, 11 labs

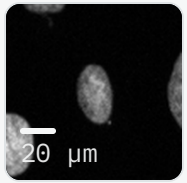
other labs

lab 5



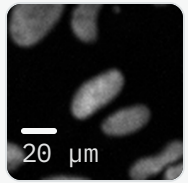
195 µm² · 0.94

lab 6



213 µm² · 0.50

lab 10



207 µm² · 0.29

AREA ≈ ≈

TEXTURE ≠ ≠

texture, not count

fig 1 93,228 wells · 11 labs · plates held out · JUMP [7]

Eleven labs. Eleven normals.

6.2

spreads apart

99.9%

told apart

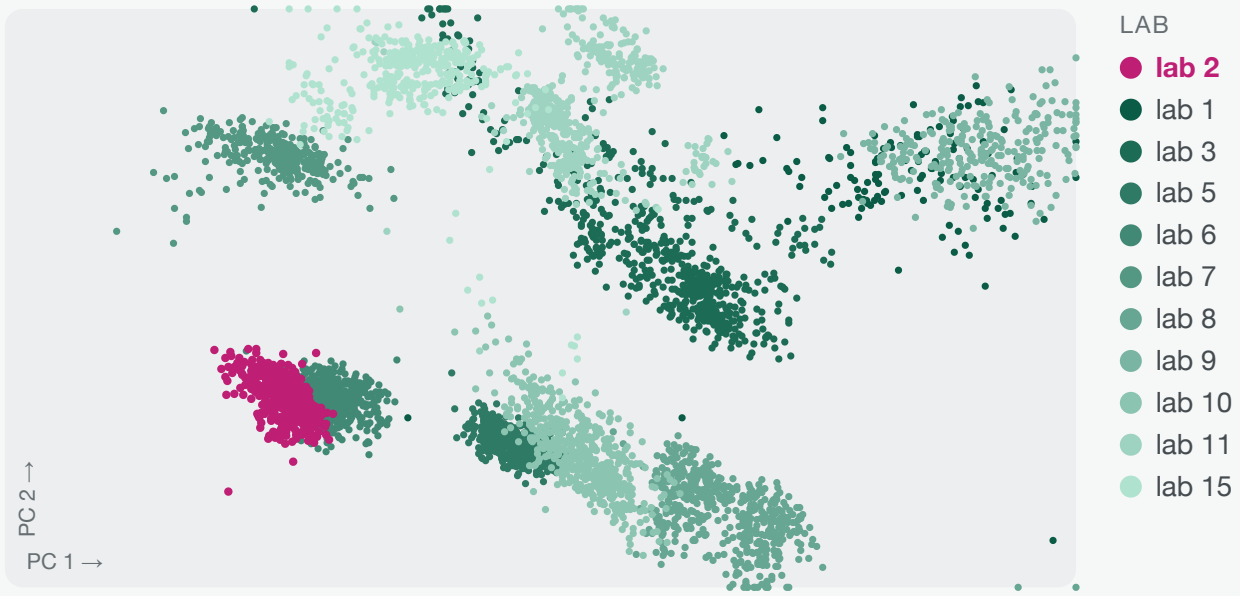
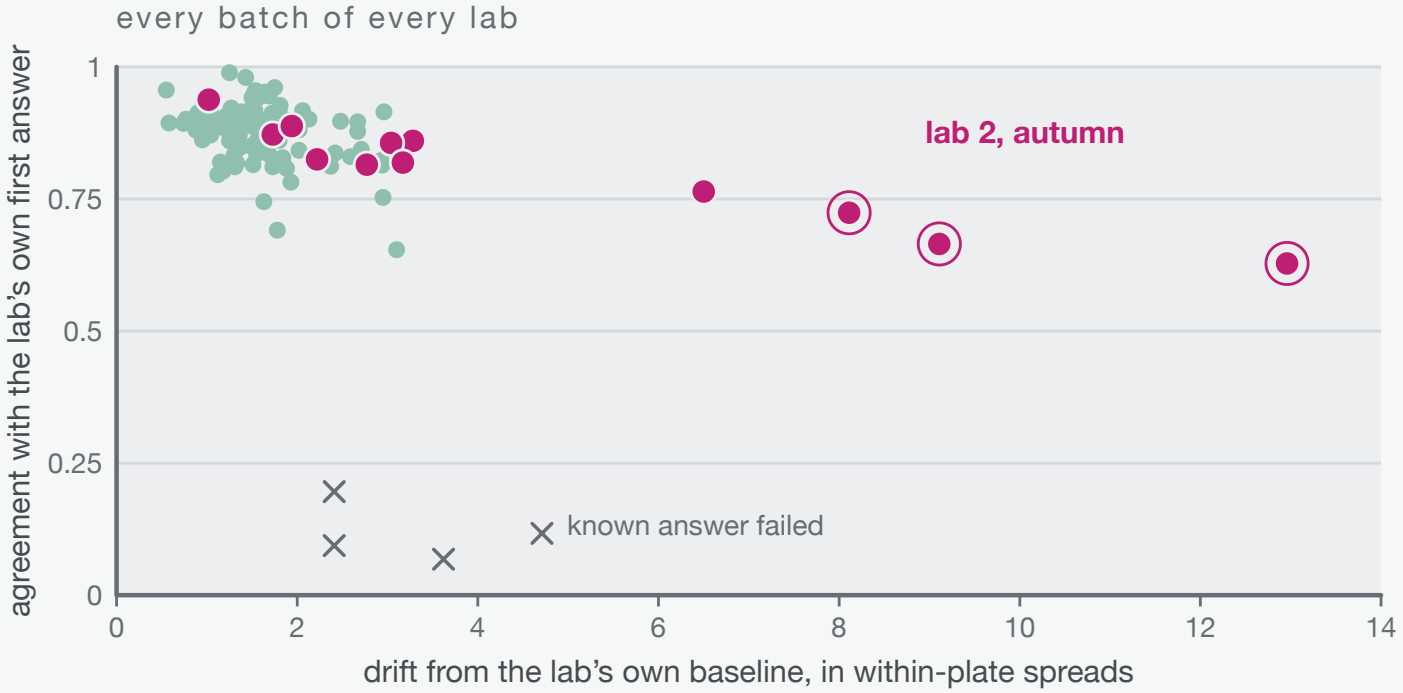


fig 2 15,352 wells · first two principal components of 3,773 numbers · JUMP [7]

click → why the map matters

Where you sit predicts what you measure.



inside labs

ρ = −0.71

drift, against your own answers

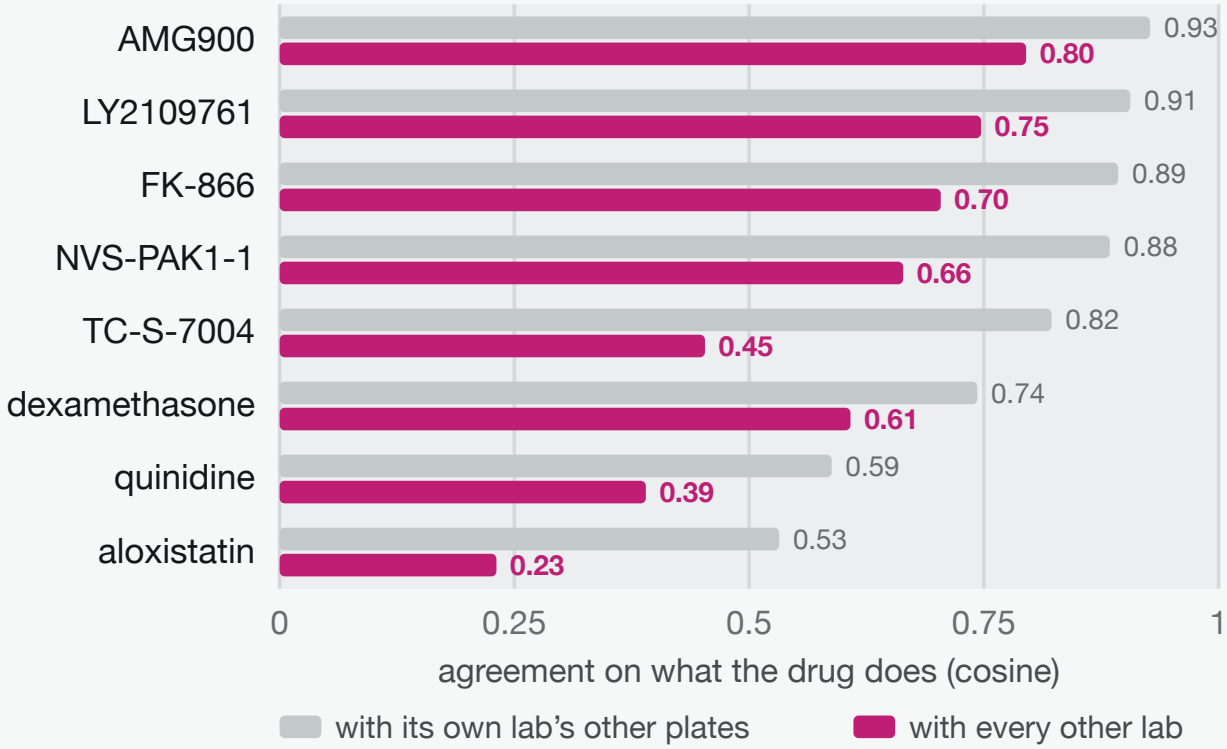
lab 2, same drugs

0.85 before

0.67 after

fig 3 one point per batch · 118 batches · JUMP [7] · A13

Does my control look like yours?



Less than it looks like your own. For all eight.

0.79 → 0.57

average agreement · own lab → other labs

ON THE MAP

0.75 → 0.78

labs agree on the same drug · 121 of 125 batches

0.52 → 0.59

drugs told apart · 125 of 125 batches

The lab comes out.
The drug stays.

fig 4 eight positive controls, run in every lab · 1,829 plates · 11 labs · JUMP [7] · A9, A14

Data is the new currency. Cell data doesn’t convert.

$1.5M a day

paid to experts for AI training data, by one vendor [10]

$30M

per cell map Roche accepts from Recursion [11]

$1.8B

pledged for virtual-cell data; funders get a year first [12]

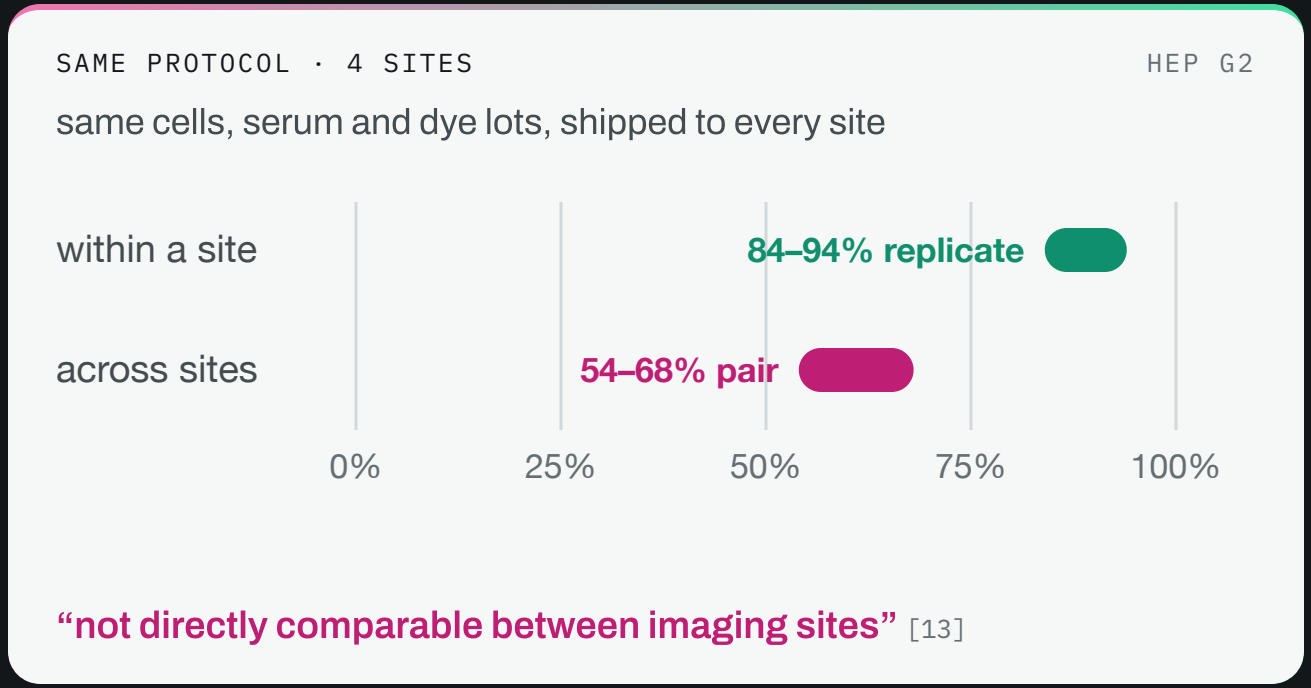
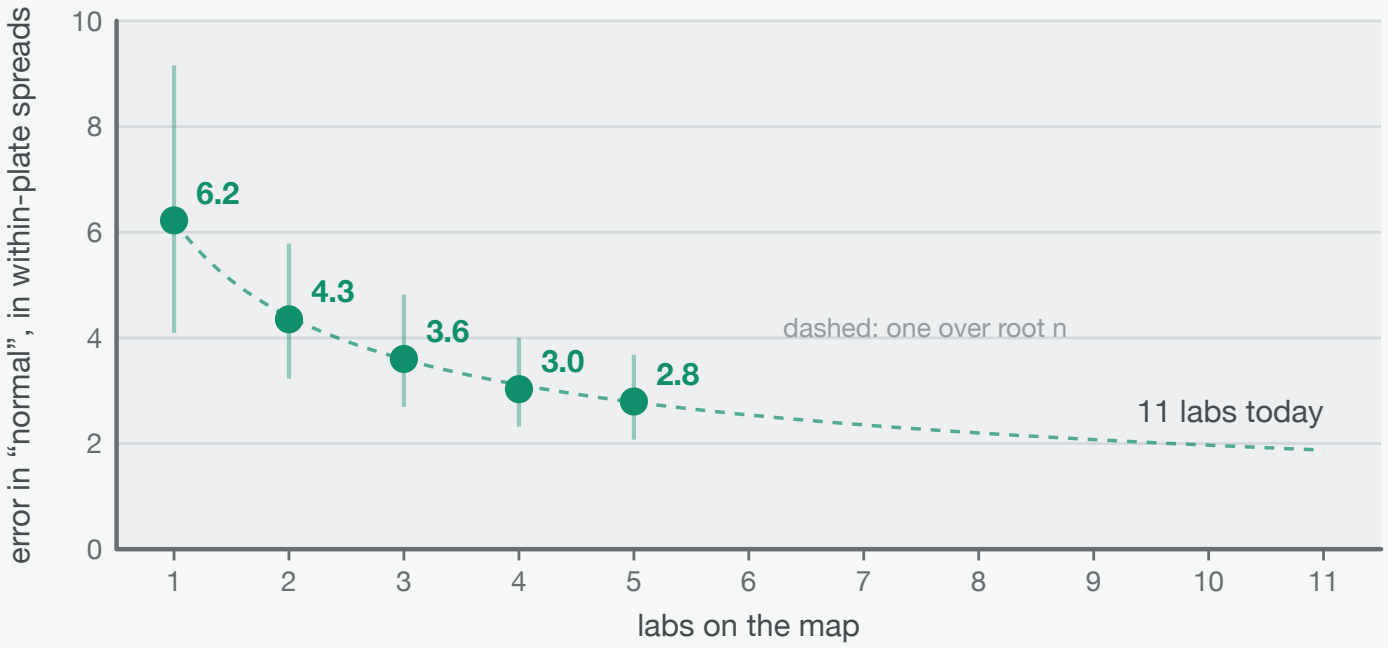


fig 5 EU-OPENSCREEN · Hep G2 · one protocol, 4 sites · as reported [13] · A23

Every lab that joins sharpens the map.

how sure the map is of normal



the map’s error

6.2 → 2.8

error in normal, one lab → five

your own baseline

0.9 → 0.4

one plate → eight

MAP open → today

CERTIFICATE paid → next

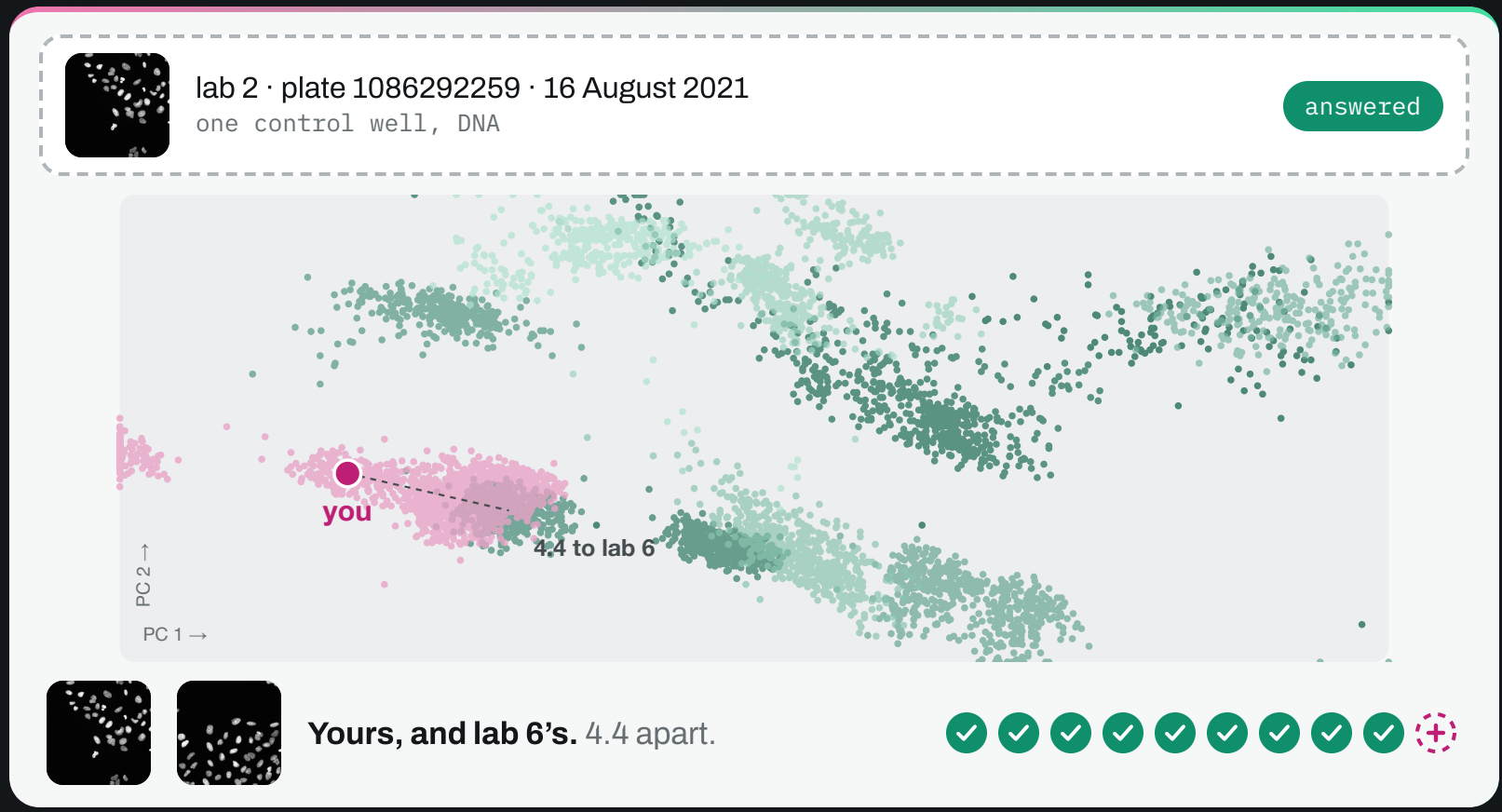
EXCHANGE a share of each trade → then

…

fig 6 two maps, different labs · 1,871 plates · JUMP [7] · A14

click → start over

Drop an image. Get an answer.



The answer

LAB 2 · 16 AUGUST 2021

CHECKED

YOUR CELLS

U2OS · 25 plates

WHERE YOU SIT

in distribution

2.8 from your first batch · 4.4 to lab 6, the nearest

YOUR OWN CHECK

188 cells per well · in range

THE MAP BEHIND IT

1,652 plates · 10 labs

YOUR DATA, MADE COMPARABLE

agreement with other labs 0.70 → 0.77

fig 7 lab 2 · 16 Aug · replayed · JUMP [7] · A11

click → the leaderboard

A game you can win.

LEADERBOARD · U2OS · 129 BATCHES · 11 LABS

controls only

RANK	LAB	BATCH	PLATES	FIELD SCORE	/100
#1	lab 5	28 Jun	7 pl		82
#2	lab 5	2 Jul	6 pl		79
#3	lab 5	6 Jul	5 pl		79
#4	lab 5	16 Jul	8 pl		78
#5	lab 5	1 Oct	6 pl		77
… 124 more batches					
#48	lab 2	2 Aug	18 pl	your best	58
#95	lab 2	16 Aug	25 pl	this batch	36

typical · tight · steady · count

Your standing

LAB 2 · 16 AUG

#95 of 129 batches · score 36

YOUR BEST

#48, 2 Aug · score 58

YOUR STREAK

9 batches in a row · the field's longest: 33, lab 5

TO CLIMB

steady: 2.9 against 1.2 for the field

THE LAB TO BEAT

lab 5 · median score 70 over 33 batches

fig 8 score = 100 − mean percentile · untreated wells · JUMP [7] · A17

click → the exchange

Your data, certified. Ready to trade.

EXCHANGE · U2OS CONTROLS · 11 DATASETS

certificate attached

DATASET	BATCHES	PLATES	CERTIFIED	OWN LAB	CONVERTS
lab 3	13	237	11 of 13	96%	0.80 → **0.81**
lab 6	13	225	13 of 13	100%	0.71 → **0.76**
lab 2	13	219	10 of 13	100%	0.75 → **0.80**
lab 10	17	205	17 of 17	100%	0.80 → **0.81**
lab 8	4	199	4 of 4	100%	0.84 → **0.88**
lab 5	33	197	33 of 33	100%	0.81 → **0.84**

… 5 more datasets

listed by the lab that made it · own lab: wells named by their cells, lowest batch · converts: agreement with other labs, as measured → on the map

Your listing

LAB 2 · U2OS · 219 PLATES

10 of 13 batches certified

CERTIFICATE

10 batches on the map · 3 flagged, 30 Aug to 3 Oct

PROVENANCE

every well named lab 2 by its own cells

CONVERTS

agreement with other labs 0.75 → 0.80 on the map

TERMS

per dataset · the lab keeps a share · proposed

fig 9 each lab's untreated wells as one dataset · replayed · JUMP [7] · terms proposed · A23, A24

Six questions, six tools.

use all of them
one did not exist

TOOL	THE QUESTION IT ANSWERS	WHAT IT DOES NOT DO
STR authentication	**Is it the line the label says?**	nothing about behaviour
vendor certificate	**Was it fine when it shipped?**	nothing after shipping
plate controls	**Did this plate work?**	cannot see a whole lab drift
phenomics platforms	**Is my own data consistent?**	removes lab differences by design
data factories	**Where do I get cell data at scale?**	one factory's normal, not every lab's
Datum Line	**Where do my cells sit, against every other lab?**	cannot say which line; fluorescence today

A7, A3, A23

Eighteen months, four milestones.

$1.8M pre-seed · A21

1

three labs on the map

U2OS on day one · any line

2

model that reads any image

99.6% today · A19

3

the benchmark playground

most-used lines · every microscope · reference perturbations · A16

4

first paid certificate

on data a company bought

then

THE EXCHANGE

certified cell data, listed by the labs that make it

THE UPSIDE

each lab keeps a share of every sale

THE RECORD

every certificate signed and timestamped in public

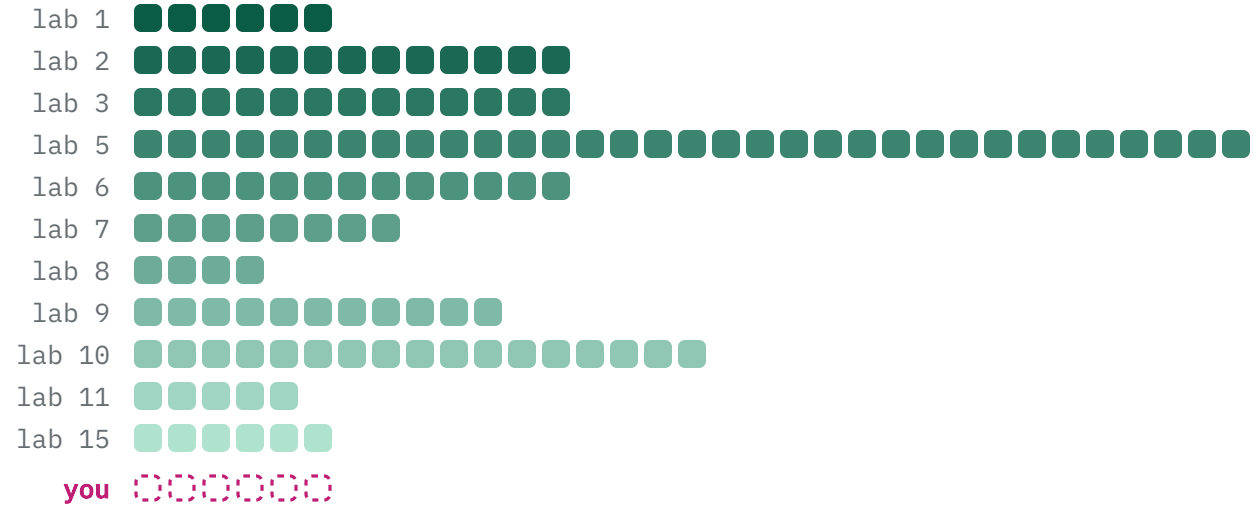
proposed · A23, A24

Every plate you image, placed on the map.

open map · signed certificates · shared upside

FIG 10 · ALREADY ON THE MAP

129 batches · 11 labs · JUMP [7]



1,871 plates · next row is yours

map today — **certificate** next — **exchange** then …

WHAT I NEED

- images
- people
- bench
- pre-seed
- partners

Have control images? Talk to me.

The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

A1 **Why not trust the plate's controls?**
There is no control well

A2 **Show me the math.**
Six operations on one embedding

A3 **Why not batch-correct it away?**
What JUMP records

A4 **Is the model the moat?**
Counting cells often wins

A5 **Why not just count cells?**
Two ways to ask one question

A6 **How often do I re-certify?**
The line is not expired

A7 **Isn't this STR?**
STR says who, images say how

A8 **Is the assay itself working?**
A known answer

A9 **Does it change the result?**
Same drug, different answer

A10 **Does it work without stains?**
Bet on brightfield

A11 **Would it cry wolf?**
Six of 129 batches

A12 **What moved?**
Stain intensity, not cell count

A13 **How does one lab move over time?**
The lab's check and the field's view

A14 **What does the field's frame buy?**
Agreement between labs

A15 **Who pays?**
Labs free; buyers of cell data pay

A16 **What would the money build?**
The ugliest dataset, on purpose

A17 **Is the leaderboard fair?**
Controls only, never results

A18 **What is private?**
Control wells leave; nothing else does

A19 **What is the machine learning?**
Image in; place, distance, move out

A20 **What about biosecurity?**
Cells cannot lie about where they came from

A21 **How much, and what for?**
Planned at $0.9M, asked at 2x

A22 **How big?**
Bottom up, every assumption on the board

A23 **What is cell data worth?**
Priced in deals; not comparable across sites

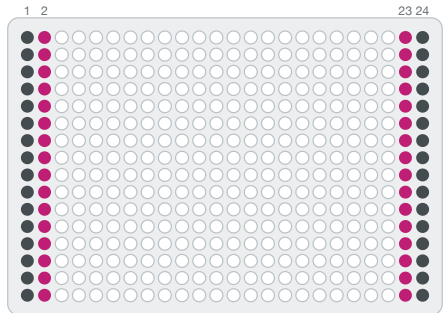
A24 **Why would a lab share?**
Owned by the lab, verified by anyone

A25 **How do labs answer it today?**
Search, ask, guess

There is no control well.

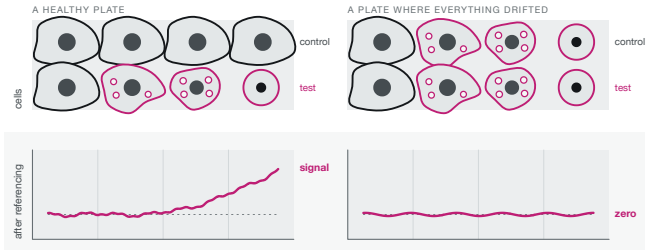
Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.

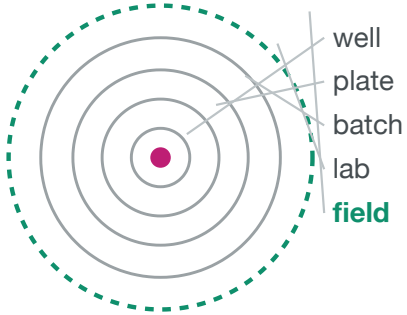
A REAL JUMP PLATE · 384 WELLS



DMSO 32 positive 32
compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.





A control only sees inside its own box.

Whole plate off: its controls move with it.
Whole batch off: so do every plate's.
Whole lab off: nobody inside it can tell.
Only the field sits outside every box.

$$x_{test} = \mu + s + \tau(s) \qquad x_{ctrl} = \mu + s \qquad \Delta = \tau(s)$$

s, the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. A1 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING **[1]**

$e = \varphi(\text{image});\ \varphi \in \{\text{CellProfiler, CNN, ViT}\}$

Which ϕ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER **99.9%**

$f : e \to \text{lab}$, held out by plate

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE **[e6]**

$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$

Difference of two means in units of the field's spread, conditioned on c. Its direction is what moved; −d is the way back.

4 · VARIANCE DECOMPOSITION **[e5]**

$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT **24%**

$e_{\perp} = e - \hat{E}[e \mid c]$, then re-run f

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS **δ**

DISTANCE $\delta = f(\|e - E[e \mid \text{vehicle}]\|)$, f bounded

CLASSIFIER $g: e' \to \text{label}$, e' aligned to the reference before comparison

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, *E[x | every variable below]*. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

99.9% raw features
balanced accuracy naming the lab

24% plate-centred
own control mean removed; chance 9%

19% plate-centred and scaled
own control spread removed too

29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

The benchmarks are so weak that counting cells often wins.

A paper I coauthored.

Accurately predicting the activity of a chemical in each bioactivity assay based on its already-known properties is extremely useful in drug development. Unfortunately, we discovered that many assays in widely used assay-activity benchmark datasets directly relate to cell health and cytotoxicity. Many other assays intend to capture a more specific phenotype, but their active compounds impact cell count, while inactives do not. In both cases, counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. To accomplish this goal, we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. Using a benchmark with 24 protein-target assays, we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. We propose several other practical recommendations for benchmarking machine learning models for predicting bioactivity and assessing the added value of mRNA, protein, or image-based profiles.

ABSTRACT, VERBATIM the finding the recommendation, and what held up

209 and 270

assays in the benchmarks the field scored itself on; prior work reported AUC above 0.9 on 32% of 209

24

protein-target assays left once cell-health assays were filtered out; there, profiles beat the count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark. That is why the benchmark is a milestone, not a paper: planted failures, a cell-count baseline on every task, every model graded (A16).

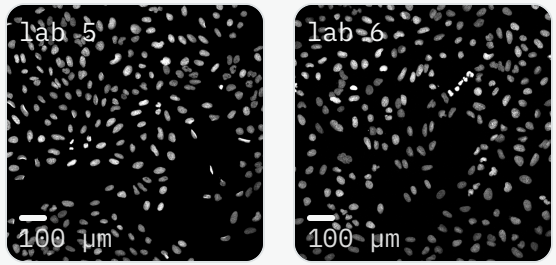
Seal S, Dee W, … Titterton K, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

PUBLISHED

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$g: e' \to \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$

Bounded, so one wild well cannot swamp the score.

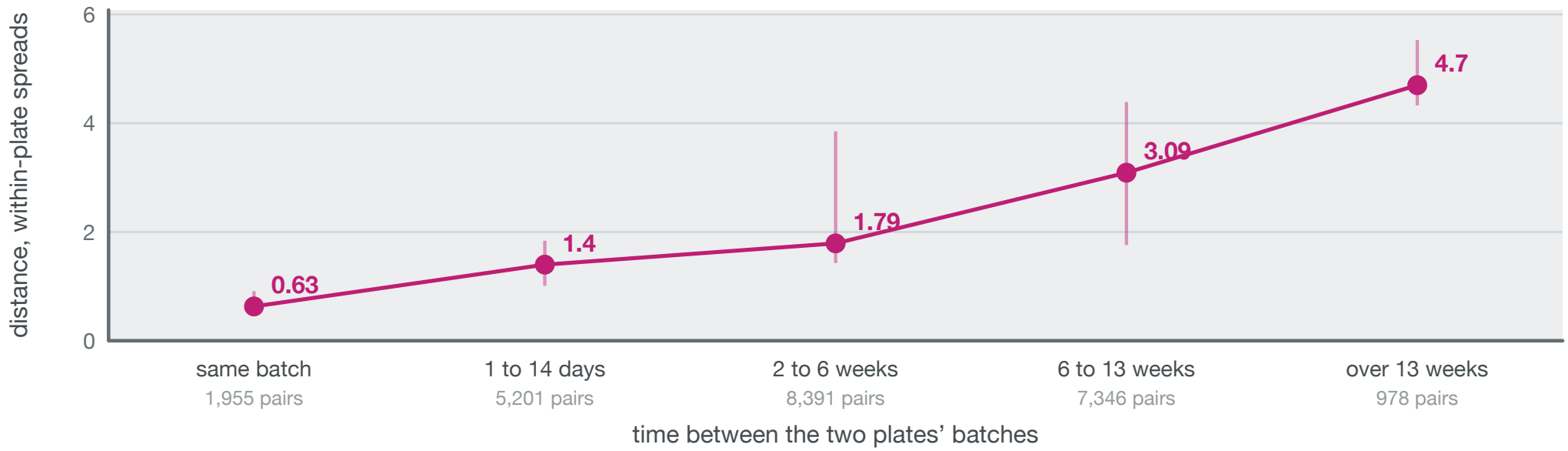
Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else's images. It takes a reference to know which.

Fig. A5 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range. results/e1b-drift.

STR says who.
Images say how.

Not a competitor: the other half of the same certificate.

STR AUTHENTICATION



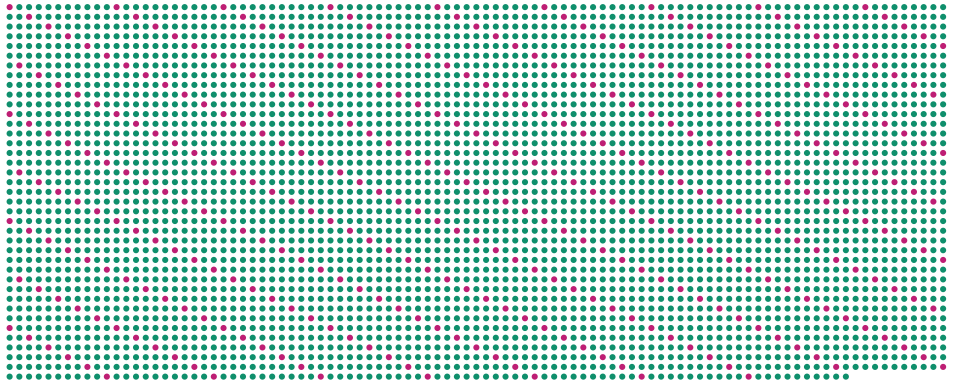
17
loci

once
per line, from a mailed card

who
which line it is

The standard for identity. Two cultures with the same fingerprint can behave differently.

DATUM LINE



3,773
numbers per well

every batch
from images already taken

how
the way it behaves, against every lab

Behaviour, placed on the field's map. Cannot say which line it is.

identity · STR + behaviour · Datum Line = CHECKED **one certificate.**

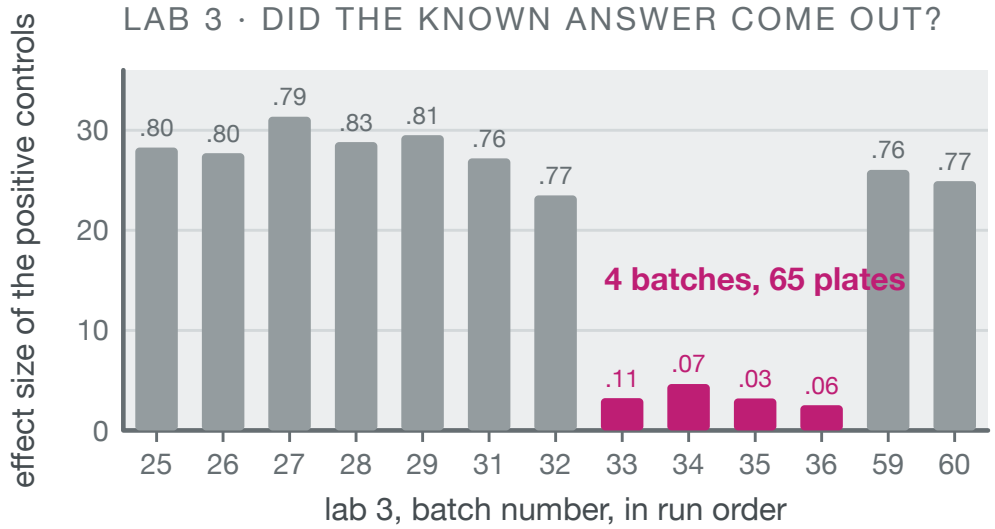
Partner, not rival: the STR provider is the natural first integration.

STR: ATCC cell authentication service, 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

A known answer, run in every lab.

Untreated wells certify where a lab sits.

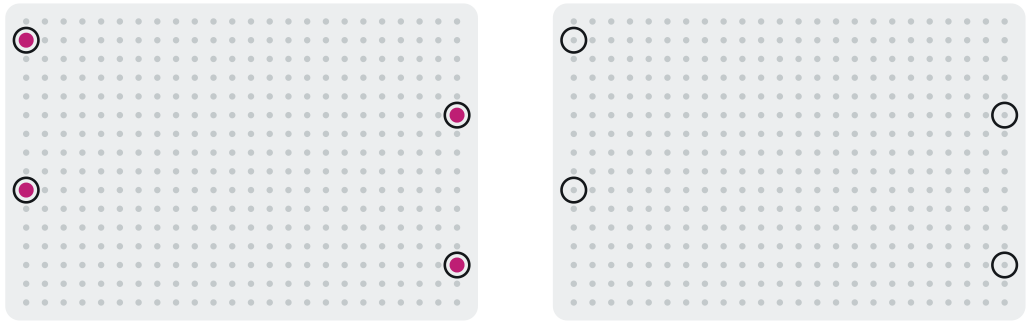
LAB 3 · DID THE KNOWN ANSWER COME OUT?



WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out
plate C13443aW

a batch that failed
plate A12069aW



○ where the plate map puts it ● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

Number above each bar: agreement with the field. 1,829 plates, JUMP [7]. Plate map: public JUMP metadata. results/e11

Same drug. Different answer.

Labs running the same assay slightly differently rank the same compounds differently.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{compound } c] - E[x \mid \text{plate } p, \text{DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.

HOW WELL A PLATE AGREES ON WHAT A DRUG DOES



Q1 · DOES IT AGREE? [E8]

Less with other labs, for every compound.

Grey: a plate's δ against the mean δ of its own lab's other plates. Pink: against each other lab's mean, averaged. AMG900: 0.93 and 0.80.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

Inside labs, yes.

Spearman between a plate's control-well distance to the other labs and its disagreement with them: 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01.

Q3 · DOES IT MATTER? [E8]

Effect size differs 14% to 79% between labs.

Effect size is |δ|. Coefficient of variation of the lab means, by compound; inside one lab, plate to plate: 10% to 44%.

Eight fixed compounds, 1,829 plates, 3,757 features, 11 labs, JUMP [7]. Z-scored on every other lab's DMSO wells, clipped at 5. results/e8-redteam

Shown in fluorescence.
Bet on brightfield.

Every number here is Cell Painting: six dyes. The product reads brightfield: no dye at all. That gap is the bet, testable on public data.

WHAT IS SHOWN

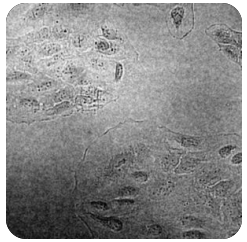
Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%. results/e1

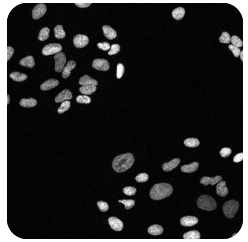
WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step: three unrecorded variables gone (A3). JUMP images brightfield beside the stains, so the same wells re-run without them.



BRIGHTFIELD



DNA STAIN

One field of one well, two ways. Lab 2, June 2021.

WHAT WOULD KILL IT

Brightfield alone at chance

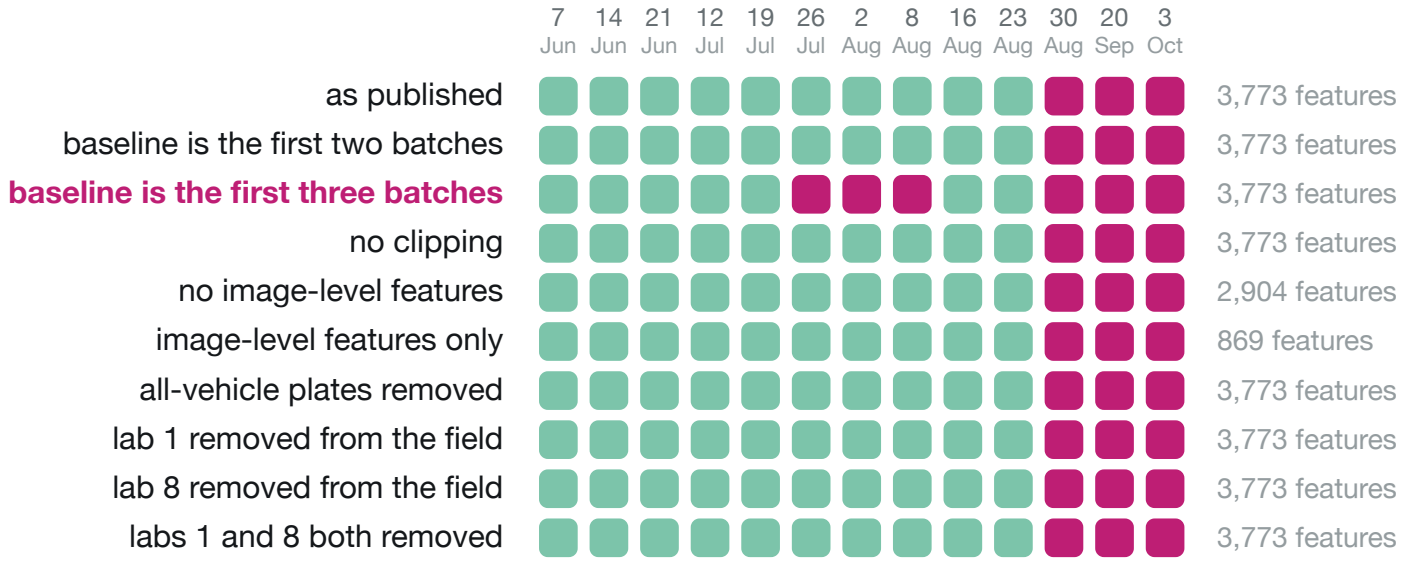
Then the certificate asks for one cheap stain. First test: a frozen vision model names the lab from one brightfield field at 99.6%, 8 labs (results/e15). **Next: the whole map from brightfield.**

Brightfield timecourse is the modality I build screens on.

Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it.

LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS



HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

WHAT CHANGES IT

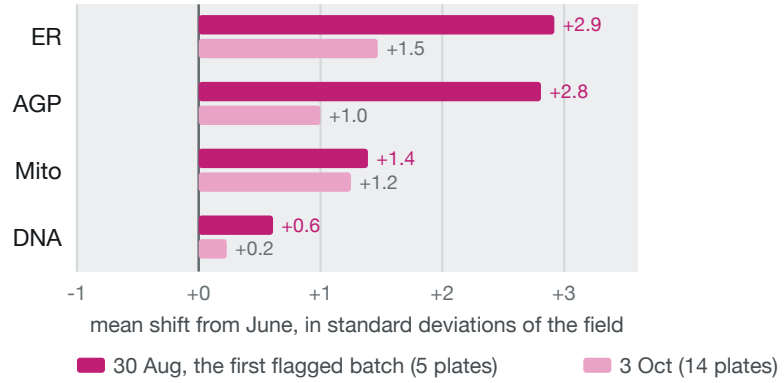
1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

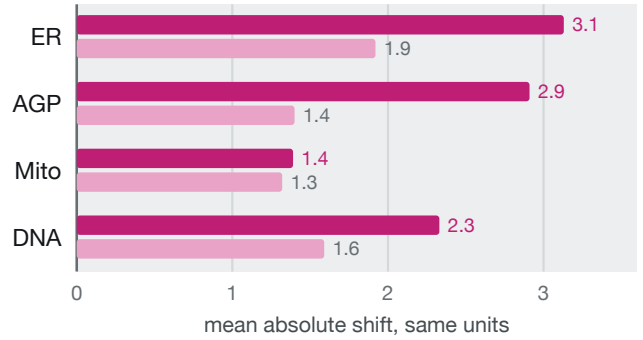
Intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.

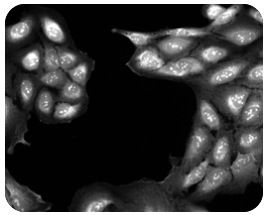
STAIN INTENSITY



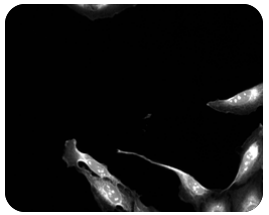
TEXTURE



AS THE MICROSCOPE SHOWS THEM



7 JUN · ER
7,612 counts



30 AUG · ER
51,230 counts

Both look fine auto-scaled. Counts 7× apart; DNA 20×.
Pattern moved too: perinuclear, then whole cell.

READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

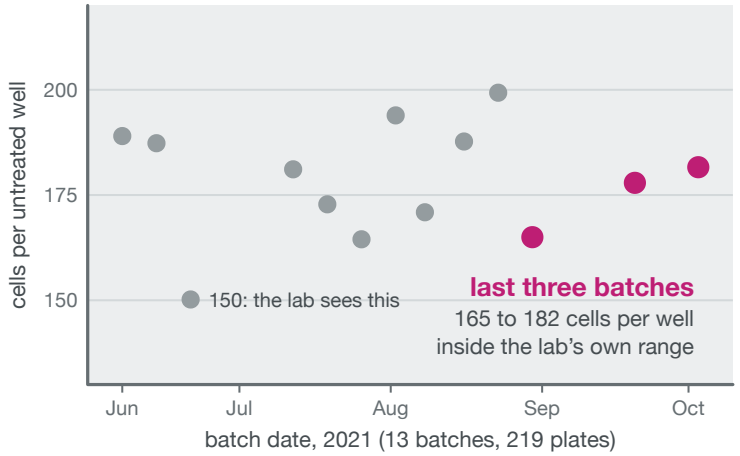
Where to look first: exposure and the ER stain, whose pattern changed as well as its brightness. Stain mix-up, bleed-through or biology are the three things to check. Not why: that is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

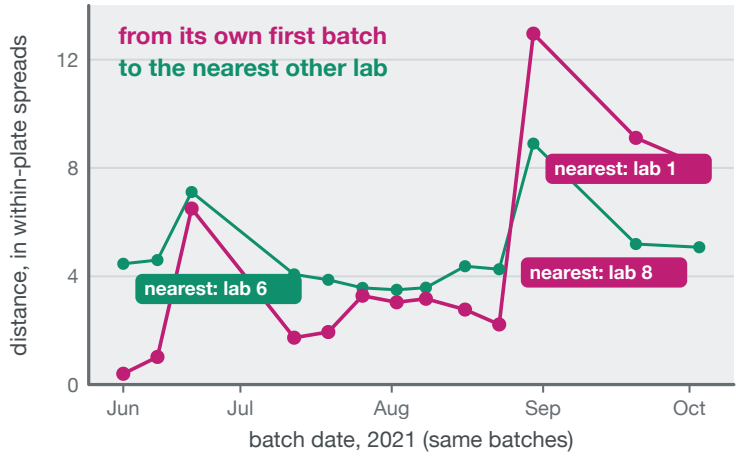
Alone, it looks fine.
Among everyone, it moves.

Same 13 batches, two views.

what the lab sees of itself



where it sits among everyone



LOOKING AT ITSELF

Cells per well stay between 150 and 199. Every batch passes. Odd things can sit comfortably inside a lab's own range.

LOOKING AT THE MAP

From 30 Aug the batches sit further from the lab's own first batch than from another lab. The lab they most resemble changes: lab 6 → lab 1 → lab 8.

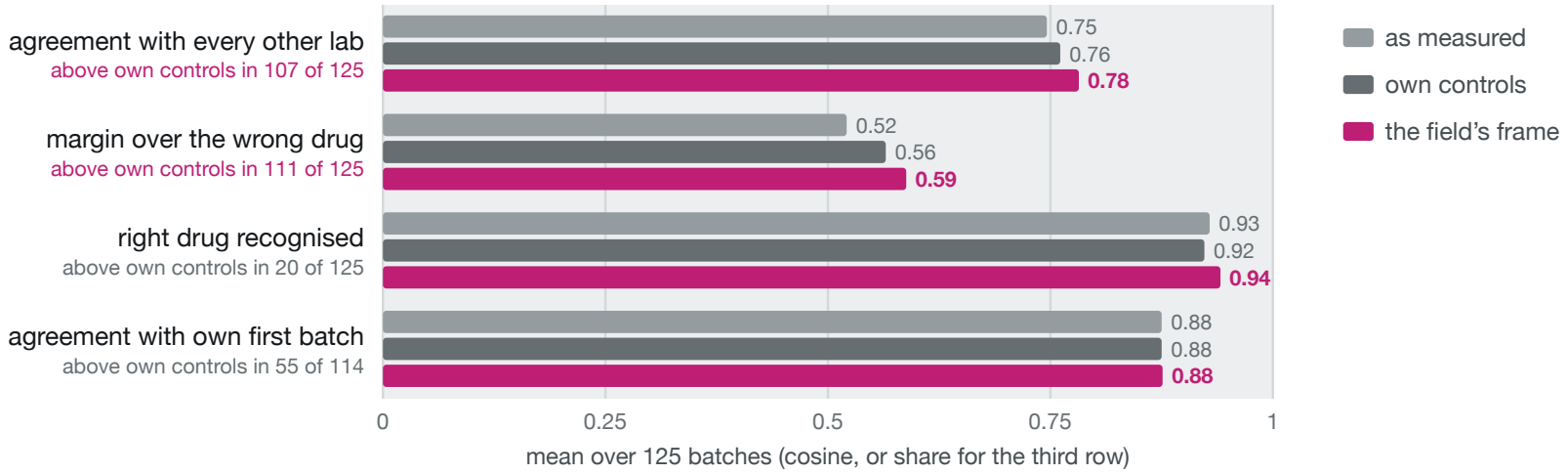
WHAT A CHANGE MEANS

Not a verdict. Resembling a different lab than you did is a reason to look: interest when you meant to change something, concern when you did not.

Distances in units of the field's within-plate spread; the nearest lab is the lab whose centre is closest. Lab 2, U2OS, 2021. JUMP [7]. What moved: A12.

One frame: what it buys.

Each batch's untreated wells are moved onto the field's, built without that lab.



DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.

1,829 plates, 11 labs, 7 positive controls, 200 principal components. Own controls: each feature standardised on the batch's untreated wells.

Labs contribute free.
Buyers of cell data pay.

The certificate pays first.
The exchange grows from it.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and cores	a place on the map, the certificate, the game	**free, for contributing controls**	the map
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	the revenue
groups that screen continuously	a reading on every batch they run	**~$5–15k per line, per quarter**	the habit
AI labs and model builders	the benchmark: models graded on planted failures	**~$50–200k a year, as partners**	the standard
then: the exchange	certified datasets, traded between labs and buyers	**a share of each trade; the lab keeps the rest**	the network

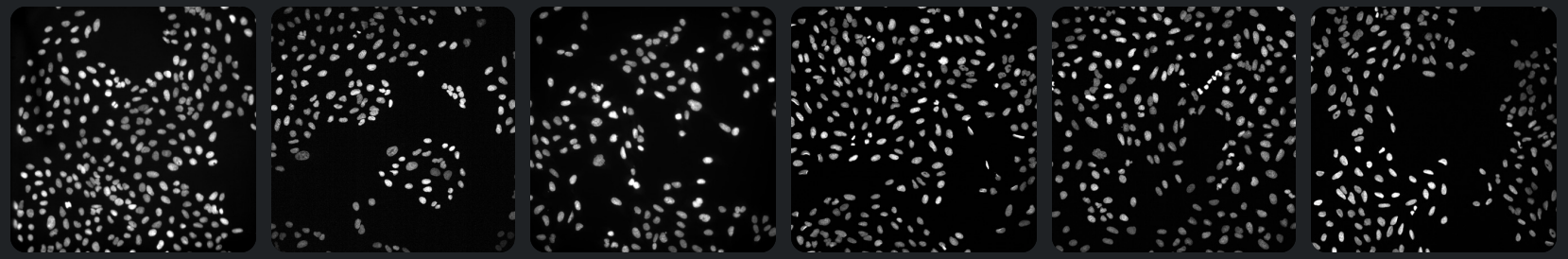
Control wells go in; readings come out; nothing else is visible to anyone. Each reading is a certificate. Prices proposed, not validated.

Build the ugliest dataset in biology, on purpose.

Same cells, same protocol, no drug. It already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

real wells, JUMP cpg0016, CC0

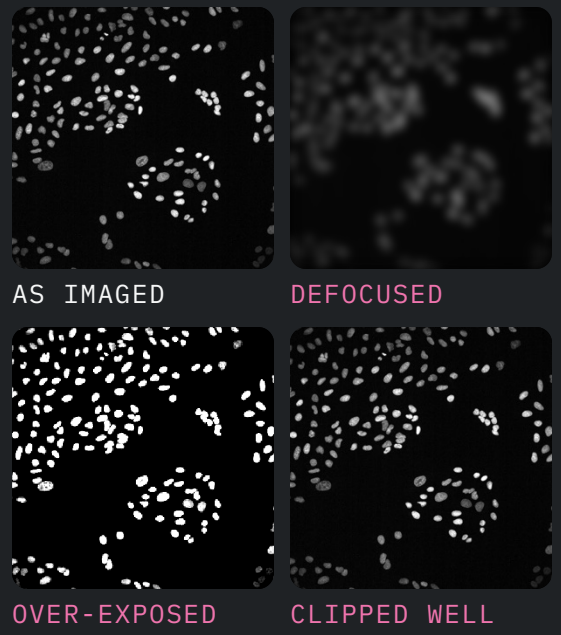


LAB 1	LAB 2	LAB 4	LAB 5	LAB 6	LAB 10
Phenix · widefield	CV8000 · confocal	Phenix · widefield	CV8000 · confocal	CV8000 · confocal	CV8000 · confocal

5	0.4–1.0	0–3	0.60–0.70	8
microscope models	objective NA	brightfield planes	µm per pixel	plate types

Across the 13 JUMP labs.

RED TEAM one well, simulated



AS IMAGED

DEFOCUSED

OVER-EXPOSED

CLIPPED WELL

A public benchmark: name the failure.

The playground: the most-used lines, every microscope, reference perturbations, planted failures. Every normalization graded.

A game, not a report card.

Every batch ranked on its untreated wells, never on results.

LAB	BATCHES	MEDIAN SCORE	BEST	STREAK	MEDIAN TYPICAL	MEDIAN TIGHT	MEDIAN STEADY	MEDIAN COUNT
lab 5	33	**70**	#1	33	3.35	0.27	0.81	0.31
lab 6	13	**60**	#16	13	4.44	0.41	1.23	0.12
lab 10	17	**54**	#10	17	3.84	0.62	1.21	0.23
lab 8	4	**52**	#54	4	4.10	1.08	0.86	0.09
lab 11	5	**47**	#23	2	3.79	0.87	1.57	0.17
lab 3	13	**44**	#18	6	3.27	0.80	2.01	0.29
lab 7	8	**42**	#69	8	6.38	0.58	1.17	0.33
lab 2	13	**39**	#48	10	5.68	0.57	1.99	0.14
lab 1	6	**32**	#78	6	8.68	1.10	2.03	0.09
lab 9	11	**25**	#97	11	9.97	0.50	1.76	0.45
lab 15	6	**23**	#68	6	5.53	0.87	1.69	0.44

THE SCORE

Four measures in within-plate spreads, each a percentile among 129 batches: distance from the field's centre, spread of the batch's plates, distance from the previous batch, cells per well against the field. 100 minus the mean.

THE BEST BATCH

lab 5, 28 Jun, 7 plates: score 82. Lab 5 holds 9 of the top ten and the longest streak, 33.

YOUR CHOICE TO PLAY

Nothing in the score touches a treated well or a result, and nobody is named without opting in. The board is for climbing; the private version is for your own plates.

1,871 plates, 11 labs, 129 batches. Streaks: consecutive batches in distribution (results/e9). JUMP [7].

Control wells leave. Nothing else does.

A control well has no compound, no target and no result, so there is nothing to clear.

WHAT LEAVES THE BUILDING

untreated control wells only

images or the well-level profile of the wells with no compound. No treated well, no compound identity, no target, no result.

a plate map of the control wells

which wells are controls; nothing about what the other wells hold.

nothing else

on this public replay that is 82,319 untreated wells from 10 other labs, and not one treated well.

WHAT DATUM LINE DOES WITH IT

builds the map

one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

answers

where the plate sits, distance from the lab's own baseline and to the nearest lab, cells per well, the verdict, the rank.

computes a move, not a model of you

the field's frame is one matrix per batch, fitted on your controls: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, 200 components.

WHAT IT NEVER SEES

your treated data

the move is sent to you and applied on your side. Your results are normalised in your building, by you.

your name, unless you choose

labs appear on the map and the board as a number by default; a lab opts in to a name. Every JUMP lab here is a number.

your images, for long

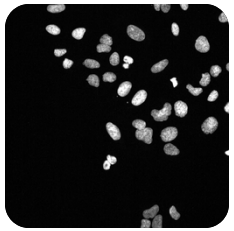
proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis in this deck runs on this rule and is public: github.com/kat-titter/datum-line/analysis. The public map is JUMP, CC0 [7]. Retention and naming are proposed, not yet contracted.

Image in.
Place, distance, move out.

Four steps. Data stays where it is; models and moves travel.

STEP 1 — AN IMAGE



One untreated well, any channel.

A Cell Painting profile today; the picture itself is now tested (step 2).

STEP 2 — NUMBERS

3,773

numbers per well today

CellProfiler today. A frozen vision model already names the lab from one picture: 99.6% brightfield, 97.2% DNA.

Next: one segmentation model, trained on every lab's nuclei.

STEP 3 — THE MAP

99.9%

a classifier names the lab; chance 9%

Z-scored on every other lab; one point per plate.

The lab being placed never sets its own scale.

STEP 4 — THE MOVE

$$x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$$

one matrix per batch, 200 components

Your batch onto the field's frame; applied on your side.

Models train the same way: they come to the data. Weights travel; treated wells never do.

PEOPLE

the box, the glimpse, the game

AGENTS

the API, the coordinates, the benchmark; models trained on every lab's controls

BOTH

every drop sharpens the map; every graded model raises the bar

One laptop runs the whole replay from cache in minutes. Every script: github.com/kat-titter/datum-line/analysis. What leaves a lab: A18.

Cells cannot lie about where they came from.

Provenance, integrity and a tripwire, from the same map.

PROVENANCE

99.9%

of 93,228 untreated wells name their true lab, on plates the classifier never saw.

A plate claimed from one lab and made in another is caught by its cells. In the replay, 119 of 129 batches have every well naming its own lab (results/e1, e9).

INTEGRITY

4 batches

65 plates whose plate map says positive control, and whose cells show no effect; no other well on those plates does either.

The data said what was done to the cells, not what the label said. Left out or mislabelled, the map cannot tell; that it is wrong, it can (results/e11).

CELLS FROM NOWHERE

9 spreads

the farthest any of 129 real batches sits from every other lab. Beyond that, a culture is not the line it claims, or not alone.

Misidentified lines and contamination are the common case; untested here, and proposed as the first partner-lab test.

SENTINELS

proposed

An untreated well is a canary. If the controls move and nobody changed anything, something in the room did: a lot, a reagent, the air, the water.

Lab 2's autumn is what that looks like on public data (A13). Alerts across labs come after the certificate.

Defensive by construction: the map holds untreated wells, the benchmark grades methods on planted imaging failures, and no result or perturbation is stored. JUMP [7].

About $1.8M for eighteen months.

People first, then the benchmark.

founder	18 months	$180k
technical co-founder	18 months	$180k
ML engineer	from month 7	$160k
payroll overhead	20% of salaries	$104k
the benchmark dataset	about 60 plates with planted failures, run where the cells already are; a CRO if needed	$120k
partner-lab bench time	three labs image the controls they already take; reagent top-ups and shipping	$30k
compute and storage	images, models, hosting, the API	$45k
legal, IP, insurance, accounting	incorporation to first contract	$45k
travel and partner visits	two benchmark partners, three labs	$20k
as planned	the lines above	$884k
the 2× rule	most things take two to three times what they should; plan at two	$884k
eighteen months	about	**$1.77M**

NO IN-HOUSE LAB

Partner labs image the controls they already take; the benchmark runs where the cells are, or at a CRO. A lab of our own would cost more and prove less.

THE UGLY DATASET IS INSIDE

$150k of the $0.9M planned, about 17%, builds the benchmark and the bench time around it. It is milestone three, not a later round.

WHAT IT HAS TO SHOW

The four milestones on slide 09. Then a seed.

Salaries are below market and assume founders are paid. The multiplier is the only line not itemised, and the one most often right. Nothing here is committed.

ASSUMPTIONS, NOT FACTS; TWO COUNTS WOULD REPLACE MOST OF THEM

About $84M a year, if everyone joined.

Bottom up.

assumption	low	base	high
organisations screening a few lines deep	200	400	800
lines each runs per quarter	2	4	8
price per line per quarter, $k	5	10	15
organisations screening panels of hundreds to thousands of lines	10	25	50
site licence per year, $k	50	150	300
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
subscriptions and site licences, per year	**$8M**	**$68M**	**$399M**
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark membership, per year	**$200k**	**$1M**	**$4M**
serviceable, per year, at full adoption	**$10M**	**$84M**	**$523M**

TWO KINDS OF LAB

Some run a few lines deep: four, say, a primary line and a co-culture. Some run thousands of genotypes wide. Per line fits the first; a site licence fits the second. Both are rows above.

THE ANCHOR

One screen run on cells that had moved is about $2.6M to redo. A certificate is a slice of that; a subscription is insurance against it.

TEN PERCENT

A tenth of the base case is about $8M a year: the five-year bar.

Prices are the proposal on A15. Academic labs and cores are free and are not in the revenue rows; they are the map.

Data has a price.
Cell data needs a standard.

Three public prices.
One experiment that shows why cell data from different sites does not add up.

AI TRAINING DATA

$1.5M a day

One vendor pays its experts more than $1.5M a day, about $85 an hour on average, at a $10B valuation. [10]

CELL MAPS

$30M

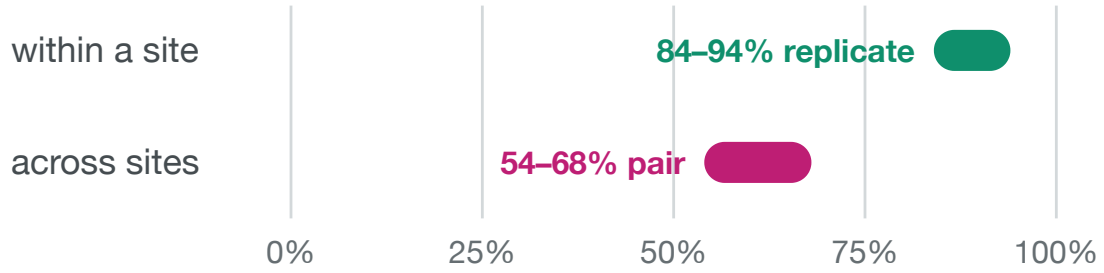
Roche and Genentech paid Recursion a second $30M milestone for an accepted whole-genome cell map; $213M from the partnership to date. [11]

CONSORTIA

$1.8B

A $1.8B virtual-cell data initiative, part new money and part existing funding; Meta, Google DeepMind and Isomorphic put in $300M together and get a year of exclusive access first. [12]

ONE PROTOCOL · 4 SITES · HEP G2



EU-OPENSCREEN ran one Cell Painting protocol at 4 sites, with cells, serum and dye lots supplied centrally. Compounds replicate inside a site far more often than they pair across sites; the authors name differences in cell culture as well as staining [13].

On the map, agreement between JUMP labs on the same drugs goes from 0.75 to 0.78 (A14). That is what makes cell data convert.

[10] TechCrunch, 2025 · [11] Recursion, 2025 · [12] eWeek, 2026 · [13] Wolff et al., *iScience*, 2025 · full references in the README

Owned by the lab. Verified by anyone.

Open science, without the token.
What a lab keeps, what anyone can check, who can buy.

CONTROL WELLS ONLY — today

Nothing else leaves a lab: no compound, no target, no result. The frame is applied on the lab's side (A18).

EVERY LAB LEAVES A FINGERPRINT — today

99.9% of untreated wells name their own lab, on plates the model never saw. A dataset claimed from one lab and made in another is caught (A20).

SIGNED AND TIMESTAMPED — proposed

Every certificate signed, its fingerprint written to a public log. Anyone can check it; nobody can backdate it. A blockchain is one such log, not a requirement.

THE LAB KEEPS ITS DATA — proposed

Listing is a choice and can be withdrawn. A share of every sale goes to the lab that made the data.

KNOWN BUYERS — proposed

Buyers verified at sign-up; unusual access flagged; every access in a log the members can read.

LOW RISK BY DESIGN — today

Images of cell lines: no human genetic data, no pathogen data. The map holds untreated wells only.

1
a lab lists
its untreated wells, with the certificate attached

2
a buyer pays
per dataset, or per line for groups that screen

3
the certificate checks
where the cells sit, who made them, what moved

4
the lab keeps a share
of every sale, for as long as the data sells

Proposed where marked; nothing is contracted. Privacy: A18. Biosecurity: A20. The open half is the map, the code and the public reference; the paid half is the certificate (A15).

Today: search, ask, guess.

WHY NOW

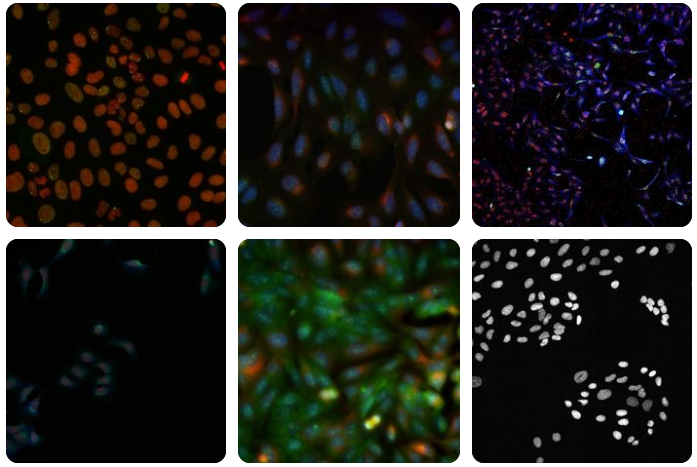
- 2023 — reference goes public [7]
- 2025 — cell data gains regulatory weight [9]
- 2026 — the map

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6] — $28B

36% ≈ $10B · reagents & references | 64% · design, analysis, protocols

1
search

do my U2OS cells look normal



2
ask around

To: whoever ran this line last

Subject: does this look right to you?

Attached: screenshot.png

3
guess

$ RUN IT ANYWAY · TOO FAST

hundreds of thousands

of R&D

HOLD · TOO SLOW

months

of development

[6] Freedman 2015 · six real U2OS papers